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import pandas as pd
from functools import reduce

# Sample data dictionary
data = {
    "CustomerID": [101, 102, 103, 104, 105],
    "Age": [25, 45, 35, 52, 23],
    "Income": [50000, 64000, 58000, 72000, 31000],
    "CreditScore": [700, 620, 680, 710, 690],
    "ExistingLoans": [0, 2, 1, 0, 3]
}

# Create DataFrame
df = pd.DataFrame(data)

# 1. Use map() to create a new column 'AgeGroup' based on Age
def age_group(age):
    if age < 30:
        return "Young"
    elif age <= 50:
        return "Middle-aged"
    else:
        return "Senior"

df["AgeGroup"] = list(map(age_group, df["Age"]))

# 2. Use filter() to find customers with CreditScore > 650 (eligible candidates)
eligible_customers = list(filter(lambda x: x > 650, df["CreditScore"]))

print("Eligible customers based on CreditScore > 650:")
print(eligible_customers)

# 3. Use reduce() to calculate total income of all customers
total_income = reduce(lambda x, y: x + y, df["Income"])
print("\nTotal Income of all customers:", total_income)

# 4. Use apply() to calculate loan eligibility based on conditions:
# Conditions: Income >= 50000, CreditScore >= 650, ExistingLoans < 2
def loan_eligibility(row):
    if row["Income"] >= 50000 and row["CreditScore"] >= 650 and row["ExistingLoans"] < 2:
        return "Eligible"
    else:
        return "Not Eligible"

df["LoanEligibility"] = df.apply(loan_eligibility, axis=1)

print("\nFinal DataFrame with Loan Eligibility:")
print(df)
```



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Search

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Eligible customers based on CreditScore > 650:
[700, 680, 710, 690]
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Total Income of all customers: 275000
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Final DataFrame with Loan Eligibility:
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	CustomerID	Age	Income	...	ExistingLoans	AgeGroup	LoanEligibility
0	101	25	50000	...	0	Young	Eligible
1	102	45	64000	...	2	Middle-aged	Not Eligible
2	103	35	58000	...	1	Middle-aged	Eligible
3	104	52	72000	...	0	Senior	Eligible
4	105	23	31000	...	3	Young	Not Eligible

```
[5 rows x 7 columns]
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